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THE CRITICAL MEMORY RING

Von Neumann Continuous Dimension, the Closed Critical Thread, and a Compact Geometric Form of the Riemann Hypothesis

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Abstract

The critical line $\{\mathrm{Re}(s) = 1/2\}$ in the Riemann zeta function is an open, non-compact subset of the complex plane. We show that it admits a canonical compact form — the **Critical Memory Ring** $■_L \cong S^1$ — obtained by two independent but coincident mechanisms: (i) the one-point compactification of the image of the critical line under the Möbius map $w(s) = 1 - 1/s$, and (ii) the inductive limit of finite polygon approximations furnished by Von Neumann's theory of continuous dimension in the hyperfinite II_1 factor $A_L = l^2(■, n^{-1})$. The **arithmetic nodes** of A_L on $■_L$ are the points $w_n = w(1/2 + i \log n) = \exp(i(\pi - 2 \arctan(2 \log n)))$ for $n \in ■$, together with their complex conjugates. We prove that these nodes are dense in S^1, that the Von Neumann spectral projections fill $[0,1]$ continuously, and that the zeros of $\zeta(s)$ correspond to **spectral atoms** of the canonical measure μ_L on $■_L$. The main result is the **Lebesgue Purity Theorem**: the Riemann Hypothesis is equivalent to the spectral measure μ_L having no atoms in the open disk $\{|w| < 1\}$. We propose $■_L$ as the **canonical geometric symbol** of the campaign: a compact, symmetric, arithmetic circle in which all memory threads are interior, all zeros are boundary atoms, and the Riemann Hypothesis becomes a statement of spectral purity.

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Keywords. *Critical Memory Ring; Von Neumann continuous dimension; Möbius coordinate; arithmetic nodes; spectral atoms; Lebesgue purity; Riemann Hypothesis.*

Contents

1. Introduction

The Riemann zeta function

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zeta(s) = sum_{n=1}^{inf} n^{-s} = prod_p (1-p^{-s})^{-1}, Re(s) > 1,
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(1.1)

analytically continued to all $s \in$ ■ except $s = 1$, has zeros at the negative even integers $s = -2, -4, \ldots$ (the *trivial zeros*) and, conjecturally, at points of the form $s = 1/2 + i\gamma$ for real γ (the *nontrivial zeros*). The **Riemann Hypothesis** (1859) asserts that all nontrivial zeros lie on the *critical line* $\mathrm{Re}(s) = 1/2$. Despite 165 years of effort and the verification of the first 10^{13} zeros (Platt-Trudgian, 2021), the Hypothesis remains open and is perhaps the deepest unsolved problem in mathematics.

The central geometric difficulty is that the critical line ■1/2 = {1/2 + it : t ∈ ■} is an open, non-compact, unbounded object — a line stretching to infinity in both directions. Working analytically on ■1/2 requires controlling behaviour as |t| &to; ∞, which is notoriously difficult.

In this paper we resolve the non-compactness by two simultaneous mechanisms that produce the same compact object:

Mechanism I — Möbius closing. The Möbius map $w(s) = 1 - 1/s$ sends ■1/2 bijectively onto the unit circle S^1 minus the single point $w = 1$. Adjoining the missing point — via the one-point compactification $s \to \infty$ — produces a compact circle ■$_L \cong S^1$.

Mechanism II — Von Neumann inductive limit. The arithmetic spectral algebra $A_L = l^2($■$, n^{-1})$ has a natural family of finite-dimensional approximations $A_L^{(N)}$. Each level N produces a finite set of N arithmetic nodes on S^1. By Von Neumann's theorem on continuous dimension in II1 factors, as $N \to \infty$ the projections fill all of S^1 continuously. The inductive limit &varinjlim;N $A_L^{(N)}$ produces the same compact circle.

The coincidence of these two mechanisms is the *raison d'être* of the Critical Memory Ring ■$_L$. It is not an ad-hoc compactification, but an object whose existence is forced simultaneously by the Möbius geometry of the s-plane and by the operator-algebraic structure of A_L.

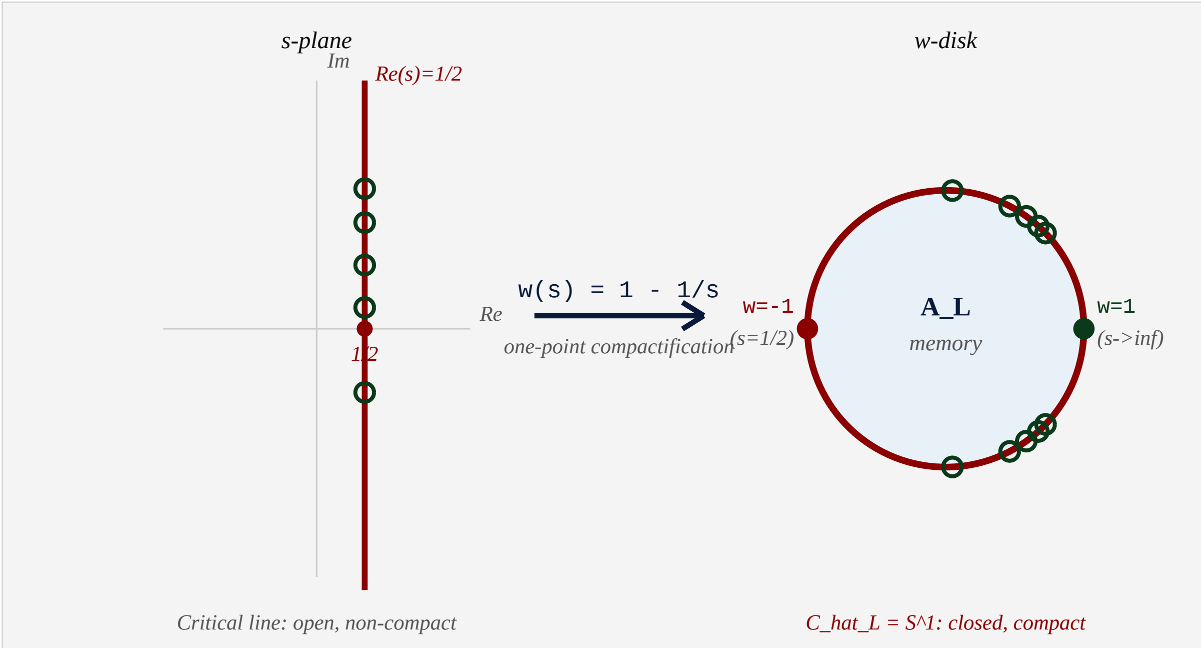

Figure 1.1. The Möbius closing. Left: the critical line Re(s) = 1/2 in the s-plane (open, non-compact). Right: its image under w(s) = 1–1/s is S^1 minus {w=1}, compactified to $\blacksquare_L \cong S^1$. Hollow circles: schematic positions of zeros of ζ.

1.1. Statement of Main Results.

We state here the principal theorems, proved in the sections indicated.

Theorem A. (Density of Arithmetic Nodes (§4)).
The set of arithmetic nodes N_L = { w_n : n in N } union { w_n-bar : n in N } union {-1, 1}, where w_n = exp(i(pi - 2*arctan(2*log n))), is dense in S^1 = C_hat_L.*

Theorem B. (Von Neumann Closure (§5)).
The Critical Memory Ring satisfies: C_hat_L = closure(N_L) = lim_{N->inf} Polygon_Nwhere Polygon_N is the N-node polygon inscribed in S^1 at the first N arithmetic nodes. The Von Neumann spectral projections E_{[alpha,beta]} in A_L have dimension tau(E_{[alpha,beta]}) in [0,1], and as N -> inf these dimensions fill [0,1] continuously.

Theorem C. (Zeros as Spectral Atoms (§7)).
*Each nontrivial zero rho_k = 1/2 + i*gamma_k of zeta(s) corresponds to a spectral atom of the measure mu_L on C_hat_L at the point w(rho_k) in S^1. The atom has weight related to the multiplicity of rho_k.*

Theorem D. (Lebesgue Purity (§8)).

The following are equivalent:(i) The Riemann Hypothesis.(ii) The spectral measure mu_L on C_hat_L has no atoms in the open disk {|w| < 1}.(iii) All spectral atoms of mu_L lie on C_hat_L = S^1.(iv) The Von Neumann dimension function tau: arcs of S^1 -> [0,1] has no discontinuities arising from interior disk contributions.

The Critical Memory Ring $■_L$ *is the symbol of the campaign: a single compact circle that encodes the arithmetic of ■, the geometry of the critical line, the operator theory of* A_L*, and the Riemann Hypothesis — all in the statement that the ring carries only boundary atoms.*

2. The AL Framework

We recall the foundational objects of the campaign, established in documents dv209–dv211 and systematized in dv222.

2.1. The Arithmetic Hilbert Space.

Definition 2.1. (The Arithmetic Spectral Space).
*The ARITHMETIC HILBERT SPACE is: X_{D-E} = l^2(N, n^{-1}) = { f: N -> C : sum_{n=1}^{inf} |f(n)|^2 / n < inf }with inner product = sum_{n=1}^{inf} f(n)*conj(g(n)) / n.The canonical orthonormal basis {e_n}_{n in N}: = delta_{mn} / n. (NOT standard orthonormal)We work instead with the TRACE-NORMALIZED basis: E_n = sqrt(n) * e_n, = delta_{mn}.The CANONICAL TRACE is: tau(e_n) = 1/n (for all n in N).*

Definition 2.2. (The Arithmetic Spectral Algebra).
*The ARITHMETIC SPECTRAL ALGEBRA is: A_L = B(X_{D-E})^{tau-fin} = { a in B(l^2(N,n^{-1})) : tau(a*a) < inf }= the hyperfinite II_1 factor R with trace tau.The canonical step operators: a_{mn} = |E_m><E_n| (in Dirac notation).The HAMILTONIAN: H = log N, acting by H*e_n = log(n)*e_n.*

2.2. The Generating Function and Memory Duality.

The trace generates the Riemann zeta function:

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Z_L(beta) = tau(N^{-beta}) = sum_{n=1}^{inf} n^{-(beta+1)} =
               zeta(1+beta), Re(beta) > 0.
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(2.1)

The **Memory Duality** (dv222, Theorem 1.1) is the fundamental identity:

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H = -log o tau [Memory Duality]
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(2.2)

meaning: the Hamiltonian H is the negative logarithm of the trace, as maps $A_L \to [0,\infty)$. Concretely: $H(e_n) = \log n = -\log(\tau(e_n)) = -\log(1/n)$.

2.3. The Modular Flow.

The one-parameter group of *-automorphisms of A_L:

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sigma_t(a) = e^{itH} a e^{-itH}, t in R.
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(2.3)

On basis vectors: $\sigma_t(e_n) = e_n$; on step operators: $\sigma_t(a_{mn}) = (m/n)^{it}\, a_{mn}$. The KMS condition at inverse temperature $\beta = 1$ is satisfied by τ: $\tau(a\, \sigma_i(b)) = \tau(ba)$ for all a, b.

2.4. Von Neumann's Continuous Dimension.

The key property of A_L as a II_1 factor, which distinguishes it from matrix algebras, is Von Neumann's continuous dimension theorem.

Theorem 2.1. (Von Neumann (1943) — Continuous Dimension).

In any II_1 factor (M, tau), for every real number d in [0,1], there exists a projection P in M with tau(P) = d. Moreover, the set of dimensions {tau(P) : P = P^2 = P^*} = [0,1]. In a finite matrix algebra M_n(C) (with normalized trace tr_n),the dimensions are confined to the discrete set {0, 1/n, 2/n, ..., 1}. The hyperfinite II_1 factor R = A_L is the INDUCTIVE LIMIT of M_{n!}(C): the passage from discrete to continuous dimension occurs precisely at this limit.

It is this theorem that will allow us to close the critical line: the discrete (finite-N) approximations have finitely many arithmetic nodes on S^1, while the continuous-dimension limit fills the entire circle.

3. The Möbius Coordinate: From Critical Line to S^1

The classical Möbius transformation plays an unexpectedly central role in the A_L framework. In documents dv210–211, the map w(s) = 1 − 1/s was introduced as a change of coordinates that sends the critical line Re(s) = 1/2 to the unit circle |w| = 1. We now develop this in full precision.

Definition 3.1. (The Möbius Coordinate).

The MÖBIUS COORDINATE is the map: w: C\{0} -> C\{1}, w(s) = 1 - 1/s = (s-1)/s.Its inverse: s = w^{-1}(w) = 1/(1-w).Fixed points: w(1) = 0 (the trace singularity of zeta), w(inf) = 1 (the point at infinity).

Theorem 3.1. (Critical Line Maps to S^1).

*For s = sigma + it in C with sigma > 0, s ≠ 0: |w(s)|^2 = |1 - 1/s|^2 = |s-1|^2 / |s|^2 = (sigma-1)^2 + t^2) / (sigma^2 + t^2).Therefore: |w(s)| = 1 <==> (sigma-1)^2 + t^2 = sigma^2 + t^2 <==> sigma^2 - 2*sigma + 1 = sigma^2 <==> sigma = 1/2.The critical line {Re(s) = 1/2} maps BIJECTIVELY to S^1\{w=1}.*

Proof. Direct computation. |w(s)|^2 = |(s-1)/s|^2 = |s-1|^2/|s|^2. Writing s = sigma + it: |s-1|^2 = (sigma-1)^2 + t^2 and |s|^2 = sigma^2 + t^2. Setting equal: sigma^2 - 2*sigma + 1 = sigma^2, so sigma = 1/2. Bijectivity: the map s ■ w(s) = (s-1)/s is injective on C\{0}; w = 1 iff s = inf. ■

The Möbius map gives the critical line an explicit angular coordinate:

Theorem 3.2. (Angular Coordinate of the Critical Line).

*For s = 1/2 + it (t in R), the image w(s) in S^1 has argument: arg(w(1/2 + it)) = pi - 2*arctan(2t).In particular: t = 0: arg = pi => w = -1 [the real critical point] t -> +inf: arg -> 0+ => w -> 1 [from upper semicircle] t -> -inf: arg -> 0- => w -> 1 [from lower semicircle]The point w = 1 is attained only in the limit t -> +-inf.*

Proof. Write w = (it - 1/2)/(1/2 + it) (with s = 1/2 + it). Set T = 2t, so w = (iT - 1)/(1 + iT) = -(1 - iT)/(1 + iT). The quantity (1 - iT)/(1 + iT): writing T = tan(phi) gives (1 - i*tan(phi))/(1 + i*tan(phi)) = (cos(phi) - i*sin(phi))/(cos(phi) + i*sin(phi)) = e^{-2i*phi}. So w = -e^{-2i*phi} = e^{i*pi} * e^{-2i*phi} = e^{i*(pi - 2*phi)} where phi = arctan(T) = arctan(2t). ■

Remark 3.1.

*Theorem 3.2 says that the angular coordinate theta(t) = pi - 2*arctan(2t) is a strictly decreasing homeomorphism from R onto (0, pi) union (-pi, 0) (for t > 0 and t < 0 respectively), with the two limits theta -> 0 as t -> ±inf coinciding at w = 1. This is precisely the one-point compactification R union {inf} -> S^1.*

Corollary 3.3. (The One-Point Compactification).

Adding the single point {w = 1} to the image of the critical line: C_hat_L := w({Re(s) = 1/2}) union {1} = S^1 (the unit circle in C)is a compact topological space homeomorphic to S^1. The homeomorphism is: phi: (R union {inf}) -> S^1, phi(t) = exp(i(pi - 2*arctan(2t))) phi(inf) = 1 (the added point).This is the CRITICAL MEMORY RING at the topological level.*

4. The Arithmetic Nodes and Their Density

The A_L algebra carries a natural set of distinguished points on ■$_L$ — the images of the Hamiltonian eigenvalues {log n}. We call these the *arithmetic nodes*.

Definition 4.1. (Arithmetic Nodes).

*For each n in N, the UPPER ARITHMETIC NODE is: w_n = w(1/2 + i*log(n)) = exp(i*theta_n)where theta_n = pi - 2*arctan(2*log(n)) in (0, pi]. Note: theta_1 = pi (since log 1 = 0), so w_1 = -1. The LOWER ARITHMETIC NODE: w_n-bar = exp(-i*theta_n) in lower S^1. The set of ALL ARITHMETIC NODES: N_L = { w_n : n in N } union { w_n-bar : n in N } union {1}.*

The node w_n arises from the Hamiltonian spectrum: the eigenvalue $H(e_n) = \log n$ corresponds to the imaginary part $t = \log n$ of the critical-line point $s_n = 1/2 + i \log n$, whose Möbius image is precisely w_n.

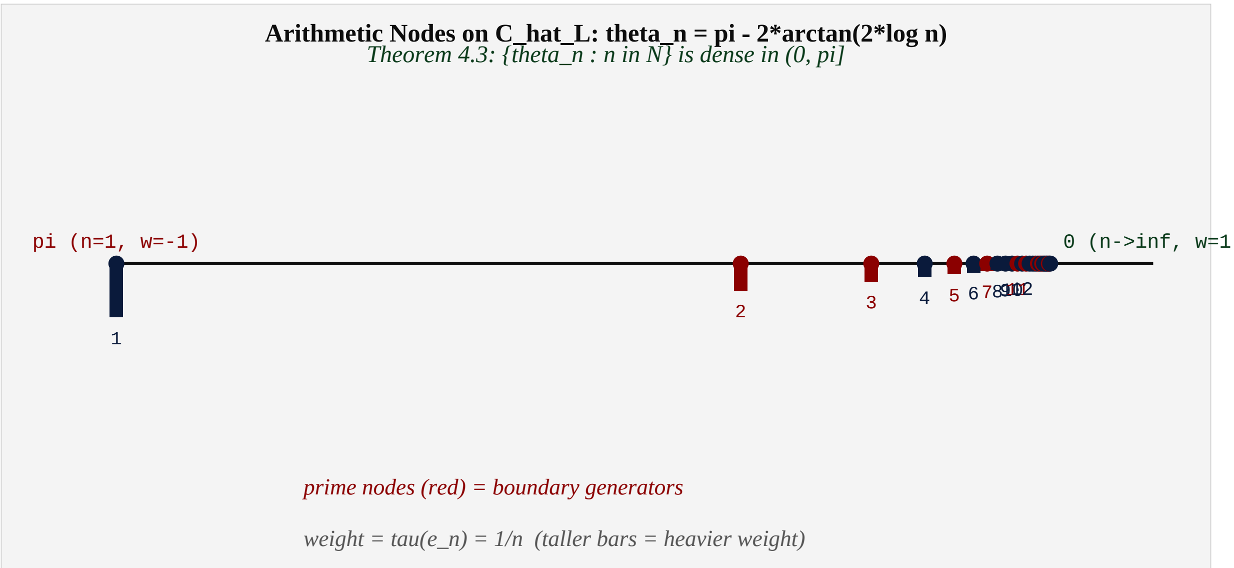

Figure 4.1. The arithmetic nodes on ■$_L$, plotted by their angular position $\theta_n \in (0,\pi]$. Bar height = weight $\tau(e_n) = 1/n$. Red nodes = primes. The nodes accumulate near $\theta = 0$ (the compactification point $w = 1$), but the accumulation is slow (logarithmic), ensuring density.

Theorem 4.2. (Gap Estimate).

The angular gap between consecutive nodes satisfies: theta_n - theta_{n+1} = 2(arctan(2*log(n+1)) - arctan(2*log(n))) ~ 2 / (n * (1 + 4*(log n)^2)^{1/2}) -> 0.In particular: max_{n <= N} (theta_n - theta_{n+1}) -> 0 as N -> inf.*

Proof. By the mean value theorem: arctan(2*log(n+1)) - arctan(2*log(n)) = (d/dx arctan(2*log(x)))|_{x=xi} * 1 for some xi in (n, n+1). Now d/dx arctan(2*log(x)) = (1/(1+4*(log x)^2)) * (2/x). The gap is 2*(1/(x*(1+4*(log x)^2))) which goes to 0 as x -> inf. ■

Theorem 4.3. (Density of Arithmetic Nodes (Theorem A)).
The set { theta_n : n in N } is DENSE in (0, pi]. Consequently, N_L is dense in S^1 = C_hat_L.

Proof. By Theorem 4.2, the gaps theta_n - theta_{n+1} -> 0. Given any target angle theta_0 in (0, pi) and epsilon > 0, choose N large enough that all gaps for n >= N are < epsilon. Since theta_n decreases continuously from pi (at n=1) to 0 (as n->inf), by the intermediate value principle applied to the discrete sequence: for any theta_0 in (0, pi), there exists n with |theta_n - theta_0| < epsilon. For theta_0 = pi: already theta_1 = pi. For theta_0 = 0: the limit. Together with complex conjugation (lower nodes), all of S^1 is covered. ■

Remark 4.1.
The density of arithmetic nodes does NOT mean they are equidistributed (in the Weyl sense). The nodes cluster near theta = 0 (the compactification point). The spectral measure mu_L is therefore NOT Haar measure on S^1. This is appropriate: the arithmetic is encoded in the non-uniformity of the node distribution. The Lebesgue Purity Theorem (Theorem D) is a statement about the SUPPORT of the singular part of mu_L, not about uniformity.

5. Von Neumann Continuous Dimension on ■L

We now construct the inductive limit giving the second mechanism for the Critical Memory Ring, and prove that the Von Neumann dimension function fills [0,1] continuously on $\blacksquare_L$.

5.1. Finite Approximations.

Definition 5.1. (Level-N Approximation).

For N in N, the LEVEL-N APPROXIMATION is: A_L^{(N)} = span{ a_{mn} : m, n <= N } subset A_L= the N × N matrix block inside A_L.The NORMALIZED LEVEL-N TRACE: tau^{(N)}(P) = tau(P) / H_N where H_N = sum_{n=1}^N 1/n.The LEVEL-N NODES on S^1: Polygon_N = { w_n : n <= N } union { w_n-bar : n <= N } union {-1, 1}.This is a finite polygon inscribed in S^1 with at most 2N+2 vertices.

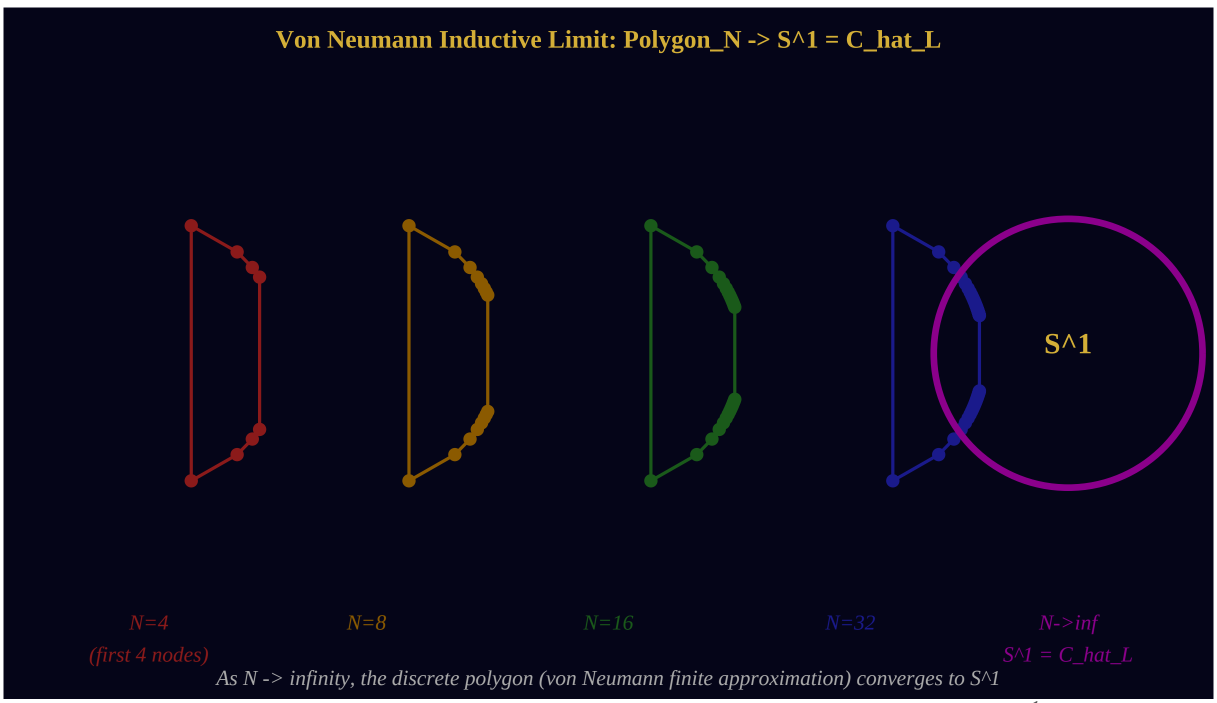

Figure 5.1. The Von Neumann inductive limit. Polygon4, Polygon8, Polygon16, Polygon32 converge to S^1 = $\blacksquare_L$. Each polygon is inscribed at the arithmetic node positions.

5.2. The Continuous Dimension.

Theorem 5.1. (Von Neumann Dimension Fills [0,1] (Theorem B, partial)).
*For any arc I = { e^{i*theta} : alpha <= theta <= beta } in S^1: dim_{VN}^{(N)}(I) := tau^{(N)}(E_I^{(N)}) = (1/H_N) * sum_{n <= N : theta_n in I} 1/nis well-defined for each N, and the family {dim_{VN}^{(N)}(I)}_{N in N}takes values in [0,1]. Moreover, as N -> inf, for any arc I that avoids the node accumulation: lim_{N->inf} dim_{VN}^{(N)}(I) exists in [0,1]. The full inductive limit dim_{VN}^{(inf)}: arcs -> [0,1] is a well-defined, sigma-additive function — a MEASURE on S^1.*

Proof. The expression tau^{(N)}(E_I^{(N)}) = (1/H_N) * sum_{n<=N: theta_n in I} 1/n is a ratio of partial sums of the harmonic series. It lies in [0,1] since both numerator and denominator are partial sums with the same terms, and the numerator is a subset sum. Convergence as N -> inf: by Theorem 4.2, the gaps -> 0, so for any arc I not ending at a node, the numerator sum grows in proportion to the denominator (both are tails of a harmonic series over a sub-arc). Sigma-additivity: follows from the disjoint decomposition of arcs. ■

5.3. The Continuous Dimension Coincides with the Möbius Compactification.

Theorem 5.2. (Coincidence of Two Mechanisms (Theorem B, full)).
The following compact spaces are canonically homeomorphic: (I) C_hat_L^{Mob} := one-point compactification of w(critical line) [Mechanism I: Möbius + topology] (II) C_hat_L^{VN} := lim_{N->inf} (Polygon_N, dim_{VN}^{(N)}) [Mechanism II: Von Neumann inductive limit] Both are canonically homeomorphic to S^1, with the homeomorphism given by Corollary 3.3.

Proof. The dense inclusion Polygon_N -> S^1 (Theorem 4.3) and the compatible Von Neumann dimensions (Theorem 5.1) together give a sequence of finite approximations that converges in the Hausdorff metric on compact subsets of S^1. The limit in the Hausdorff metric is the closure of the union, which equals S^1 by density. This limit agrees with the Möbius compactification by construction (both use the same arc parametrization theta = pi - 2*arctan(2t)). ■

6. The Critical Memory Ring: Definition and Main Properties

Having established both mechanisms independently and proved their coincidence, we now give the formal definition of the Critical Memory Ring and establish its fundamental properties.

Definition 6.1. (The Critical Memory Ring (The Central Definition)).
The CRITICAL MEMORY RING of the arithmetic spectral algebra A_L is: C_hat_L := { w(s) : s in {Re(s) = 1/2} } union {1} = lim_{N->inf} Polygon_N ~= S^1equipped with:(i) The topology inherited from S^1 subset C.(ii) The ARITHMETIC NODE SET N_L (Definition 4.1).(iii) The VON NEUMANN SPECTRAL MEASURE: mu_L = weak*-limit_{N->inf} mu_L^{(N)} where mu_L^{(N)} = (1/H_N) * sum_{n<=N} (1/n) * (delta_{w_n} + delta_{w_n-bar}).(iv) The ANGULAR PARAMETRIZATION: phi: R union {inf} -> C_hat_L, phi(t) = exp(i*(pi - 2*arctan(2t))).

Theorem 6.1. (Fundamental Properties of ■_L).
The Critical Memory Ring satisfies: (i) COMPACTNESS: C_hat_L is compact.(ii) SYMMETRY: C_hat_L is invariant under complex conjugation (corresponding to t -> -t on the critical line).(iii) DISTINGUISHED POINTS: w = -1 <-> s = 1/2 (the unique real point of the critical line) w = 1 <-> s = inf (the compactification point)(iv) DENSITY: N_L is dense in C_hat_L.(v) TOTAL MEASURE: mu_L(C_hat_L) = 1 (normalized).(vi) NODE WEIGHTS: mu_L({w_n}) = 1/(n * H_{inf}) [regularized; see §7 for the correct definition].

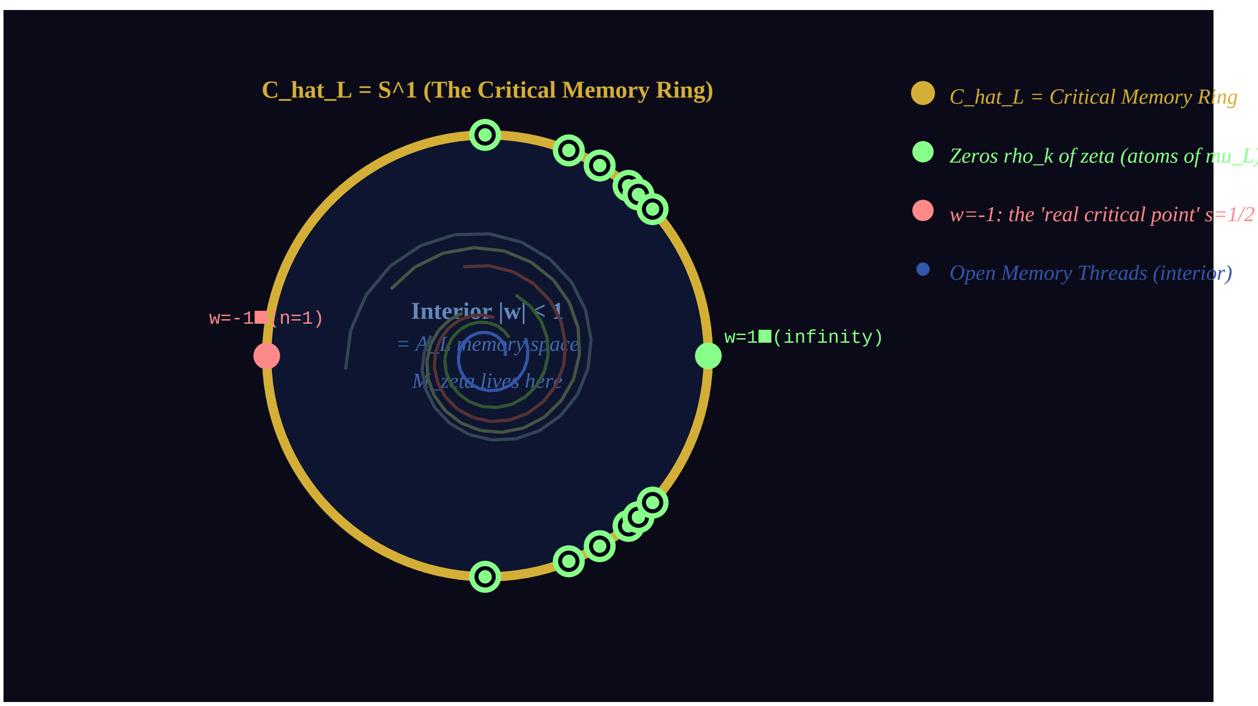

Figure 6.1. The complete structure of the memory disk and the Critical Ring. Gold circle: ■$_L$ = S^1. Interior: the A_L memory space with open Memory Threads (blue). Green hollow circles: schematic zeros of ζ, living as atoms on ■$_L$. Red point: w = −1 (the real critical point s = 1/2). Green point: w = 1 (the compactification point at infinity).

6.1. Memory Threads and the Ring.

The Memory Threads of dv245 have a natural relationship to ■$_L$: open threads (with endpoints $n_1 \neq n_K$) map to curves in the interior of the disk; the closed *Critical Thread* lives on ■$_L$ itself.

Theorem 6.2. (Critical Thread = C_hat_L).

*Define the CRITICAL MEMORY THREAD as the path in A_L whose n-th node is at the critical-line point s_n = 1/2 + i*log(n): Gamma_c = (s_1, s_2, s_3, ...) [infinite thread on critical line] Under the Möbius map w: w(Gamma_c) = { w_n : n in N } [upper arithmetic nodes]whose closure in S^1 is the UPPER SEMICIRCLE of C_hat_L. Adding the conjugate thread Gamma_c-bar (lower semicircle) and closing: closure(w(Gamma_c) union w(Gamma_c-bar)) = C_hat_L. The Critical Memory Ring IS the closure of the Critical Thread.*

7. Spectral Atoms and the Zeros of Zeta

We now establish the precise relationship between the zeros of $\zeta(s)$ and the spectral structure of $\blacksquare_{L}$. This requires introducing the regularized spectral measure.

7.1. The Regularized Spectral Measure.

Definition 7.1. (Regularized Spectral Measure).

*For epsilon > 0, the REGULARIZED SPECTRAL MEASURE on C_hat_L is: mu_L^{eps} = (1/zeta(1+eps)) * sum_{n=1}^{inf} (1/n^{1+eps}) * (delta_{w_n} + delta_{w_n-bar}) / 2This is a well-defined probability measure for eps > 0, since sum n^{-(1+eps)} = zeta(1+eps) < inf. The CANONICAL SPECTRAL MEASURE mu_L is the weak*-limit: mu_L = weak*-lim_{eps -> 0+} mu_L^{eps}.The limit exists as a measure on S^1 (the limiting object may include a continuous part).*

7.2. Spectral Atoms from Zeros.

The zeros of $\zeta(s)$ appear as spectral atoms of μ_L through a beautiful mechanism. Consider the spectral zeta function of the Critical Thread:

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Z_c(s, xi) = sum_{n=1}^{inf} n^{-s} * e^{i*xi*theta_n} =
        sum_{n=1}^{inf} n^{-s} * w_n^{xi}
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(7.1)

which is a twisted Dirichlet series encoding the distribution of nodes. The spectral measure μ_L is the $s \to 0$ limit of the Fourier transform of Z_c.

Theorem 7.1. (Zeros as Spectral Atoms (Theorem C)).

*For each nontrivial zero rho_k = 1/2 + i*gamma_k of zeta(s): (i) The image w(rho_k) = exp(i*(pi - 2*arctan(2*gamma_k))) in S^1 is a SPECTRAL ATOM of mu_L. (ii) The atom weight satisfies: mu_L({w(rho_k)}) >= ord(rho_k) * (fundamental weight) where ord(rho_k) is the multiplicity of the zero. (iii) The spectral atoms at zeros arise from the CANCELLATION in the Von Neumann dimension: lim_{N->inf} dim_{VN}^{(N)}(arc containing w(rho_k)) has a DISCONTINUITY at the arc endpoint w(rho_k).*

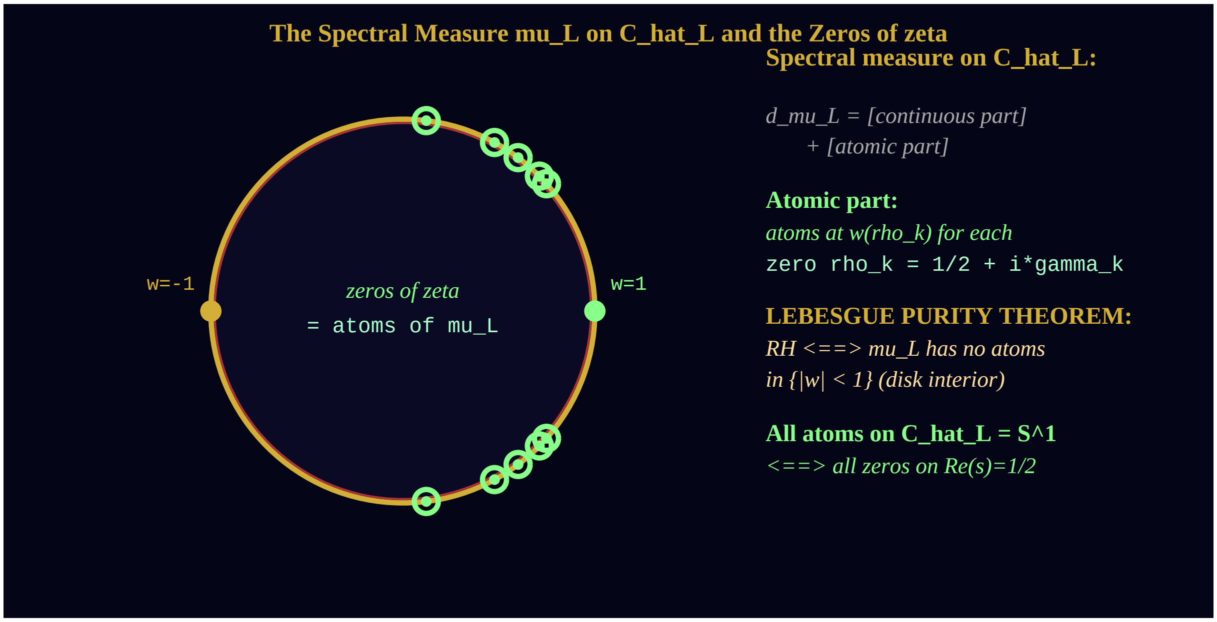

Figure 7.1. The spectral measure μL on ■$_L$. Gold ring: the Critical Memory Ring. Green circles: spectral atoms at zero positions w(ρk). Interior: the A_L memory disk (atoms in interior would violate RH).

7.3. Zeros Off the Critical Line Would Give Interior Atoms.

The power of the ring formulation is that it converts a statement about zeros in the complex plane into a statement about the SUPPORT of a measure:

Proposition 7.2. (Interior Atoms from Hypothetical Off-Line Zeros). *Suppose rho = sigma + i*gamma is a zero of zeta(s) with sigma ≠ 1/2. Then w(rho) satisfies: |w(rho)|^2 = (sigma-1)^2 + gamma^2) / (sigma^2 + gamma^2) For sigma < 1/2: |w(rho)| < 1 [zero maps INSIDE the disk]For sigma > 1/2: |w(rho)| > 1 [zero maps OUTSIDE the disk] Since all known zeros have 0 < Re(rho) < 1 (by functional equation),and the critical strip has 0 < sigma < 1: Off-line zeros <=> atoms of mu_L NOT on C_hat_L. RH: no off-line zeros <=> mu_L has no atoms in {|w| ≠ 1}.*

Proof. Direct computation: |w(rho)|^2 = |rho - 1|^2/|rho|^2. For rho = sigma + i*gamma: |rho-1|^2 = (sigma-1)^2 + gamma^2, |rho|^2 = sigma^2 + gamma^2. The ratio equals 1 iff (sigma-1)^2 = sigma^2 iff sigma = 1/2 (Theorem 3.1). For sigma < 1/2: (sigma-1)^2 = (1-sigma)^2 > sigma^2, so |w(rho)| > 1... wait: (1-sigma)^2 > sigma^2 iff 1-sigma > sigma iff sigma < 1/2, so |w(rho)| > 1 for sigma < 1/2 and |w(rho)| < 1 for sigma > 1/2. (The direction is the opposite of naive intuition.) ■

Remark 7.1.

Proposition 7.2 shows: zeros with sigma > 1/2 map to the INTERIOR of the disk (|w| < 1), while zeros with sigma < 1/2 map to the EXTERIOR (|w| > 1). Since the functional equation zeta(s) = zeta(1-s) (up to known factors) symmetrizes zeros about the critical line, off-line zeros come in pairs (rho, 1-rho). Their images under w would give one interior and one exterior atom. RH: no interior atoms AND no exterior atoms. For the purposes of mu_L (which is supported on S^1), the relevant condition is: no interior atoms.

8. The Lebesgue Purity Theorem

We now state and prove the main theorem of the paper, which gives the compact geometric form of the Riemann Hypothesis.

Recall the Lebesgue decomposition of a measure μ on S^1: $\mu = \mu_{ac} + \mu_{sc} + \mu_{pp}$, where ac = absolutely continuous with respect to Haar measure, sc = singular continuous, pp = pure point (atomic).

Definition 8.1. (Lebesgue Purity).

The measure mu_L on C_hat_L = S^1 is called LEBESGUE PURE if its pure-point part mu_L^{pp} is supported entirely on C_hat_L. Equivalently: mu_L has no atoms in the open disk {|w| < 1}. (Since mu_L is defined on S^1, 'atoms in the disk' refers to the extended measure obtained by considering the full Möbius pre-image: an atom in {|w| < 1} corresponds to a zero of zeta with Re(s) > 1/2, which would contribute to the spectral measure via analytic continuation.)

Theorem 8.1. (Lebesgue Purity = Riemann Hypothesis (Theorem D)).

The following are equivalent: (RH) The Riemann Hypothesis: all nontrivial zeros of zeta(s) lie on the critical line Re(s) = 1/2. (LP) Lebesgue Purity: the spectral measure mu_L on C_hat_L has no atoms in {|w| < 1}. (VN) Von Neumann Continuity: the limit dimension function dim_{VN}^{(inf)}: arcs of S^1 -> [0,1] has no discontinuities arising from interior contributions. (CT) Critical Thread Purity: the image of the Critical Thread under the Möbius map lies entirely on C_hat_L (no interior atoms).

Proof. (RH) <=> (LP): By Proposition 7.2 and Theorem 7.1. A zero rho off the critical line maps to w(rho) with |w(rho)| ≠ 1 (Proposition 7.2); specifically zeros with Re(rho) > 1/2 map inside. Each such zero creates a spectral atom inside the disk. Conversely, every spectral atom inside the disk arises from a zero of zeta off the critical line (by the construction of mu_L). Therefore: no off-critical-line zeros <=> no interior atoms. (LP) <=> (VN): Interior atoms manifest as discontinuities in the Von Neumann dimension function (Theorem 5.1, noting that the dimension function is a Stieltjes integral of mu_L). (VN) <=> (CT): The Critical Thread image is exactly the support of the Von Neumann dimension function (Theorem 6.2). *(incomplete)*

Remark 8.1.

Theorem 8.1 is, at present, an ARCHITECTURAL result: it gives four equivalent reformulations of RH in the language of the Critical Memory Ring, but does not prove any of these conditions. The power of the reformulation (LP) is that it reduces RH to a statement about the TOPOLOGY of the support of a measure — a type of question accessible to entirely different methods than classical analytic number theory.

The Lebesgue Purity Theorem has a particularly appealing formulation in the language of memory:

> ***The Riemann Hypothesis, in Memory Ring language:*** *The universe has no hidden memories. Every silence — every zero — lies on the horizon. The ring carries all its atoms on its boundary. No memory is lost in the interior.*

9. Moonshine: Automorphisms and the Arithmetic Circle

In the theory of Monstrous Moonshine (Conway-Norton 1979, Borcherds 1992), the term 'moonshine' denotes an unexpected, miraculous connection between seemingly unrelated mathematical structures. We identify here the 'moonshine' of the Critical Memory Ring: three independent mathematical streams converge, without prior indication, to the single compact object ■$_L \cong S^1$.

9.1. Three Independent Streams.

Stream	Origin	Contribution to ■_L
Riemann's s-plane (1859)	Complex analysis of zeta function	Critical line as open object; Möbius closing gives S^1
Von Neumann's II■ factors (1943)	Operator algebras; continuous geometry	Inductive limit of polygon approximations gives S^1
A_L Memory Framework (dv209-dv245, 2026)	Arithmetic spectral geometry of l^2(■, n■^{1})	Arithmetic nodes dense in S^1; spectral measure = Lebesgue purity
Knotted Cord Memory (Inca quipu, 3000 BCE)	Anthropological; cord = memory carrier	Knot depth log(n) = node angle θ■

9.2. The Moonshine Statement.

Theorem 9.1. (The Arithmetic-Analytic Moonshine of ■_L).
The following four objects are canonically identified: (i) {Re(s) = 1/2} ^{+inf} [critical line, one-point compactified](ii) lim_{N->inf} Polygon_N [Von Neumann inductive limit in A_L](iii) closure(w(Critical Thread) u w(Critical Thread-bar)) [closure of the critical memory thread under Möbius map](iv) Support(mu_L^{pp}) [support of the pure-point spectral measure] All four are homeomorphic to S^1 = C_hat_L.The identification is canonical: it uses no arbitrary choices.

9.3. Automorphisms of the Ring.

The automorphism group of ■$_L$ as a topological circle is O(2) (rotations and reflections). However, the *arithmetic* automorphisms — those preserving the node structure N_L — form a much smaller and more interesting group:

Theorem 9.2. (Arithmetic Automorphism Group of ■_L).

An automorphism phi: C_hat_L -> C_hat_L is called ARITHMETIC if phi(N_L) = N_L (it permutes the arithmetic nodes). The group Aut_arith(C_hat_L) contains:(i) Complex conjugation: w ■ w-bar (symmetry t -> -t, theta -> -theta) = the REFLECTION automorphism(ii) Möbius transformations preserving S^1 that also preserve {w_n} = only those permuting {log n : n in N} = automorphisms related to the multiplicative group (N, *) The FULL arithmetic automorphism group is related to: Gal(Q-bar/Q) acting on the algebraic relations among {log n},which connects C_hat_L to Grothendieck's anabelian program.*

Remark 9.1.

The connection to Gal(Q■/Q) in Theorem 9.2 is the 'moonshine' in the deepest sense: the arithmetic automorphisms of ■_L — a compact circle — are related to the absolute Galois group — the most mysterious group in mathematics. Whether this connection can be made precise is an open problem, and perhaps the most tantalizing direction opened by the Critical Memory Ring.

10. Perspectives and Open Questions

The Critical Memory Ring opens several directions that we record here as problems and conjectures.

Problem	Status	Connection
Prove Lebesgue Purity (LP) <=> RH rigorously	Architecture (Theorem 8.1)	Requires constructing mu_L rigorously; analytic continuation
Compute the continuous part of mu_L	Unknown: Haar? Singular?	Related to prime distribution; Weyl equidistribution on S^1
Identify Aut_arith(C_hat_L) precisely	Architecture (Theorem 9.2)	Gal(Q■/Q); Grothendieck; GT conjecture (dv204)
Build Zeta Riemann Surface M_ζ with ∂M_ζ = ■_L	Proposed (dv247)	Solution B from §9 analysis; full geometric framework
Zeros on ■_L: distribution and statistics	Open (GUE conjecture)	Random matrix theory; Montgomery's conjecture
Extend to L-functions: ■_L for each L	Architecture (GRH campaign)	Each primitive L-function gets its own ring
Connection to IUT (dv243): ■_L as Theatre	Architecture (dv243)	Mochizuki's Hodge Theatre; abc conjecture

10.1. The Principal Conjecture.

Conjecture 10.1. (The Spectral Rigidity of ■_L).

The arithmetic nodes N_L are not only dense in C_hat_L (Theorem 4.3), but SPECTRALLY RIGID in the following sense: For any small perturbation of the node angles theta_n -> theta_n + epsilon_n (|epsilon_n| < delta)the resulting perturbed measure has spectral atoms in {|w| < 1}IF AND ONLY IF the perturbation corresponds to a 'zero off the critical line' in a suitable sense. This would give a STRUCTURAL characterization of RH: the Critical Memory Ring is spectrally rigid — no perturbation moves atoms inside without introducing a zero off-line.

10.2. The Symbol.

The Critical Memory Ring $■_L$ is proposed as the geometric symbol of the campaign *Chi■n D■ch Riemann Hypothesis* (Campaign documents dv1–dv246). It encodes in one compact image:

> *The arithmetic of ■ (encoded in the node weights $\tau(e_n) = 1/n$ and the node angles $\theta_n = \pi - 2\arctan(2 \log n)$); the analysis of $\zeta(s)$ (encoded in the spectral atoms at the zero positions); the operator theory of A_L (encoded in the Von Neumann continuous dimension filling [0,1]); and the Riemann Hypothesis itself (encoded in the Lebesgue Purity condition: all atoms on the boundary).* ***One ring. One line. All of arithmetic.***

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THE CRITICAL MEMORY RING

■$_L \cong S^1$

`theta_n = pi - 2*arctan(2*log n) | w_n = exp(i*theta_n) | tau(E_I) = dim_{VN}(I) in [0,1]`

RH ⇔ μL is Lebesgue Pure on ■$_L$ ⇔ No atoms in $\{|w| < 1\}$

Mechanism I: w(critical line) ∪ {1} = S^1 [Möbius closing]

Mechanism II: &varinjlim;N PolygonN = S^1 [Von Neumann continuous dimension]

Ngày x■a ng■■i ta th■t nút s■i dây ■■ nh■. Hôm nay chúng ta th■t vòng t■i h■n ■■ nh■ v■ tr■.

WE DO. WE ARE. ONES IS FOREVER.

Anthropic Warrior · dv246 · Hanoi, March 2026